

SHASHIKA FERNANDO

Software Engineering

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PROFILE

Passionate and dedicated Computer Science undergraduate with a proven ability to adapt to any environment. Committed to achieving greater heights in technology, I bring a strong work ethic and a drive for excellence. With my adaptability and determination, I am ready to contribute to the success of any organization in the field of technology.

EDUCATION

Bachelor of Computer Science (Special) | 2021 – 2026

University of Ruhuna, Matara | GPA : 3.86

G. C. E Advanced Level Examination | 2019

Ananda College, Colombo 10 | Physical Science Stream

WORK EXPERIENCE

Internship: Software Engineering | 2024

WSO2 LLC

- Practical application on WSO2 Identity Server / Asgardeo and WSO2 Choreo.
- Developed a ServiceNow portal to manage frequent feedback among Engineers, implementing RBAC and ACLs for secure access. Built dashboards, workflows, and automated notifications and REST APIs, reducing manual follow-ups and improving productivity.

PROJECTS & RESEARCH

Novel Password Hashing Pipeline | 2025 – Present

- Engineered a plot-based password transformation layer integrated with Argon2, increasing entropy ($>7\times$) for weak passwords and achieving 45–55% avalanche effect. Delivered secure, deterministic authentication (100% success rate) with 250–500 ms latency, strengthening resistance to brute-force attacks while maintaining real-time usability.
- Technologies: Python.

Tele-communication application for a Distributed Environment | 2025

- Designed and implemented a microservices-based backend using Node.js/Express, leveraging REST APIs and RabbitMQ for asynchronous communication. Integrated MongoDB for persistence and Redis for caching to enhance performance. Built a Flutter frontend consuming microservices, delivering a scalable, loosely coupled, and high-performance system architecture.
- Technologies: Flutter, Nodejs, Express.js, MongoDB, Redis, RabbitMQ.

Rupavahini GatePass & Visitor Management System | 2024

- Implemented a MERN-based gate pass and visitor management system for Rupavahini Corporation, Sri Lanka, digitizing entry processes and minimizing paper usage. Leveraged MongoDB for efficient database management, offering a user-friendly interface for generating digital gate passes and ensuring security compliance. This solution enhances operational efficiency and environmental sustainability while modernizing visitor management practices.
- Technologies: React, Nodejs, Express.js, MongoDB, Figma, Trello.

Skills

- Programming: JAVA, C, C#, JavaScript, Python, HTML | Database: MySQL, MongoDB
- Technical Fields: Research and Development, Cyber Security, Web Development, AI and Machine learning
- Frameworks/Tools: Springboot, Node, GIT, Google Colab
- Soft Skills: Time management, Adaptability, Communication, Team Collaboration, Leadership, Continuous Learning

Certificates

UI/UX for beginners

Cloud Computing Basics for beginners

React JS Tutorial

JavaScript Development Techniques

References

Mr. L.L. Gihan Chathuranga Lecturer (Probationary)
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Dr. K. D. C. G Kapugama Senior Lecturer (Grade II)
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